

1. Determine whether the integral converges and, if so, evaluate the integral.

(a) $\int_1^\infty \frac{1}{1+x^2} dx$

(g) $\int_{-\infty}^\infty \frac{1}{e^x + e^{-x}} dx$

(b) $\int_2^\infty \frac{1}{x^2-1} dx$

(h) $\int_0^1 \arcsin x dx$

(c) $\int \frac{dx}{x(\ln x)^2}$

(i) $\int_1^\infty 2^{-x^2} dx$

(d) $\int_0^\infty e^{-x} \sin x dx$

(j) $\int_e^\infty \frac{dx}{\sqrt{x+1} \ln x}$

(e) $\int_{-\infty}^0 x e^x dx$

(k) $\int_0^\infty \cos^2 x dx$

(f) $\int_1^4 \frac{1}{x^2-5x+6} dx$

(l) $\int_0^{\pi/2} \frac{\cos x}{\sqrt{\sin x}} dx$

2. (a) For what values of r is $\int_0^\infty x^r e^{-x} dx$ convergent?
 (b) Show by induction that $\int_0^n x^n e^{-x} dx = n!$, $n = 1, 2, \dots$.

3. The integral

$$\int_0^\infty \frac{1}{\sqrt{x}(1+x)} dx$$

is improper in two distinct ways: the interval of integration is unbounded and the integrand is unbounded. If we rewrite the integral as

$$\int_0^1 \frac{1}{\sqrt{x}(1+x)} dx + \int_1^\infty \frac{1}{\sqrt{x}(1+x)} dx,$$

then we have two improper integrals, the first have an unbounded integrand and the second defined on an unbounded interval. If each these integrals converges with values L_1 and L_2 , then the original integral converges and has value $L_1 + L_2$. Evaluate the original integral.

4. Show that $\int_{-\infty}^\infty \sin x dx$ diverges although $\lim_{b \rightarrow \infty} \int_{-b}^b \sin x dx = 0$.

5. The function

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$

is important in statistics. Prove that the integral on the right converges for all real x .

6. Evaluate

$$\int_0^2 \arcsin x dx$$

using integration by parts even though the technique leads to an improper integral.

7. Use comparison test to determine whether the integral converges.

(a) $\int_1^\infty \frac{x}{\sqrt{1+x^5}} dx$

(b) $\int_1^\infty 2^{-x^2} dx$

8. A nonnegative function f defined on $(-\infty, \infty)$ is called a *probability density function* if

$$\int_{-\infty}^\infty f(x) dx = 1$$

- (a) Show that the function f defined by

$$f(x) = \begin{cases} 6x/(1+3x^2)^2, & x \geq 0; \\ 0, & x < 0. \end{cases}$$

is a probability density function.

- (b) Let $k > 0$. Show that the function

$$f(x) = \begin{cases} ke^{-kx}, & x \geq 0; \\ 0, & x < 0. \end{cases} \quad (1)$$

is a probability density function. It is called the exponential density function.

- (c) The *mean* of a probability density function f is defined as the number

$$\mu = \int_{-\infty}^{\infty} xf(x)dx.$$

Calculate the mean for the exponential density function defined in (1).

- (d) The *variance* of a probability density function f is defined as the number

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx,$$

where μ is the mean. Calculate the variance for the *exponential density function* defined in (1).

9. Test these series for (a) absolute convergence, (b) conditional convergence.

(a) $\sum (-1)^k \frac{k}{\ln k}.$

(e) $\sum (-1)^k \frac{k+2}{k^2+k}.$

(b) $\sum \frac{k!}{(-2)^k}.$

(f) $\sum (-1)^{k+1} \frac{k^k}{k!}.$

(c) $\sum \sin\left(\frac{k\pi}{4}\right).$

(g) $\sum (-1)^k ke^{-k}.$

(d) $\sum \frac{(-1)^k}{k-2\sqrt{k}}.$

(h) $\sum (-1)^k \left(\sqrt{k+1} - \sqrt{k}\right).$

10. Let L be the sum of series

$$\sum_{k=0}^{\infty} (-1)^k \frac{1}{k!}$$

and let s_n be the n th partial sum. Find the least value of n for which s_n approximates L within 0.001.

11. Determine whether the series converges or diverges

(a) $\sum \frac{(\ln k)^2}{k}.$

(b) $\sum \frac{2^k k!}{k^k}.$

$$(c) \sum \frac{1}{k} \left(\frac{1}{\ln k} \right)^{3/2}.$$

$$(d) \frac{1}{2} + \frac{2}{3^2} + \frac{4}{4^3} + \frac{8}{5^4} + \cdots.$$

$$(e) \sum \frac{1}{1 + 2 \ln k}.$$

$$(f) \sum \frac{2 + \sin k}{k^2}.$$

$$(g) \sum \frac{2k}{(2k)!}.$$

$$(h) \sum k e^{-k^2}.$$

$$(i) \sum \frac{2k!}{(2k)!}.$$

$$(j) \sum \left(\frac{k+1}{k} \right)^k.$$

$$(k) \sum \frac{(-5)^k}{4^{k+1}}.$$